

128 Gb/s CAP-16 Transmission Over 100 m OM4 MMF Using Low-Complexity ABCD Linearization of a Directly Modulated VCSEL

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Abstract We demonstrate 128-Gb/s (32-Gbaud) CAP-16 transmission over 100-m OM4 MMF using a low-cost commercial 21-GHz VCSEL and low-complexity ABCD linearization. The method successfully mitigates the VCSEL non-linearity, achieving a $\sim 2\times$ EVM reduction vs conventional FFE. FFE-only implementation fails to support this data rate. ©2026 The Author(s)

Introduction

The rapid growth of cloud computing and artificial intelligence is driving strong demand for higher-capacity short-reach optical interconnects for data centres. These links must support increasing data rates while maintaining low cost and power consumption. Directly modulated lasers (DMLs), particularly vertical-cavity surface-emitting lasers (VCSELs), remain the dominant light sources for intensity-modulation direct-detection (IM/DD) short-reach optical links due to their high energy efficiency, compact footprint, and low manufacturing cost [1]. However, at high symbol rates, especially when operated beyond their 3 dB bandwidth (BW), the intrinsic non-linearity of DMLs introduces waveform distortion that limits the achievable data rates [2].

Several digital signal processing (DSP) techniques have been explored to mitigate DML non-linearities and extend the data rate. For example, 160 Gb/s discrete multitone (DMT) transmission using a 26 GHz VCSEL over 10 m OM4 multi-mode fibre (MMF) has been demonstrated using Volterra equalization [3], while lookup-table predistortion enabled 112 Gb/s pulse amplitude modulation (PAM-4) transmission with a 28 GHz VCSEL over 50 m OM5 MMF [4]. Machine-learning-based equalization has also achieved 53.6 Gbaud PAM-4 with a 22 GHz VCSEL [5]. However, these approaches rely on complex DSP, hence increasing power consumption and system complexity, which conflicts with the stringent energy-efficiency requirements of data-centre interconnects.

We previously proposed the ABCD approximation method [6–9], a simplified implementation of the full back-calculation [10] for mitigating non-linearity in DMLs. The method pre-distorts the electrical drive signal by approximating the ideal modulation current required to produce a linear optical output [6]. ABCD linearization has been experimentally demonstrated for amplitude-based formats

including non-return-to-zero (NRZ) [7] and PAM-4 [8]. Recently, we showed that the method can also linearize two-dimensional carrierless amplitude/phase (CAP) signals while preserving the orthogonality between the in-phase (I -) and quadrature (Q -) components, enabling 32 Gb/s CAP-16 transmission using a 10 Gb/s-class VCSEL [9].

Building on this proof-of-concept demonstration, this work extends ABCD linearization to a commercially mature VCSEL with a measured 3 dB bandwidth of 21 GHz (specified for 25 Gb/s NRZ operation), and experimentally demonstrates 128 Gb/s (32 Gbaud) CAP-16 data transmission over 100 m OM4 MMF. Using ABCD linearization and a 20-tap receive feed-forward equalizer (FFE), the link achieves a root-mean-square error vector magnitude (RMS EVM) of $\sim 11\%$ and an estimated bit error rate (BER) below the hard decision FEC (HD-FEC) threshold of 3.8×10^{-3} . This corresponds to a $5.1\times$ increase in data rate relative to the nominal 25 Gb/s NRZ operation for which the VCSEL was designed. The FFE-only implementation of the link fails at this data rate and reach. The results demonstrate that the low-complexity ABCD linearization can effectively mitigate VCSEL non-linearity in practical MMF links, relaxing transmitter BW requirements and enabling ≥ 100 -Gb/s-per-lane using low-cost, mature VCSEL technology.

ABCD Approximation Method

The ABCD approximation method, previously described in detail in [6–9], is briefly summarized here. The required modulation current is approximated as [6]:

$$I_{bc}(t) \approx \tilde{I}_{bc}(t) = \tilde{I}_A(t) + \tilde{I}_B(t) + I_C(t) + I_D \quad (1)$$

Here, I_D corresponds to the threshold current, while the linear component, I_C [6]:

$$I_C = \frac{qV}{\Gamma\tau_p} \cdot N_p(t) \cdot \left(1 + \frac{\varepsilon}{g_0\tau_e}\right) \quad (2)$$

is directly determined by the target photon density, $N_p(t)$ and laser parameters, where q is

the electron charge, V the active region volume, Γ the optical confinement factor, τ_p the photon lifetime, ε the gain compression factor, g_0 the gain slope coefficient, τ_e the carrier lifetime. The non-linear correction terms ($\tilde{I}_B$ and $\tilde{I}_A$) are obtained from $I_C(t)$ through [6]:

$$\tilde{I}_B(t) = b \frac{dI_C(t)}{dt}, \quad \tilde{I}_A(t) = a \frac{d\tilde{I}_B(t)}{dt} = a \cdot b \frac{d^2 I_C(t)}{dt^2} \quad (3)$$

where the scaling factors a and b depend on the laser parameters and operating bias point [6]. The method therefore requires only low-order derivatives and simple scaling operations to implement [11].

Experiment Setup

The experimental setup is shown in Fig. 1. A commercial 850 nm VCSEL (Broadcom AFCD-V51KC1) [12] with a measured ~ 21 GHz 3 dB BW was used as the transmitter (Tx). The device was electrically connected using a 40 GHz ground-signal-ground (GSG) probe. The emitted light was coupled into a MMF patchcord using $\times 16$ and $\times 18$ objective lenses. The precise values for the intrinsic device parameters of Eq. (2) are unknown, but they are estimated from the measured L-I characteristics, and S_{21} frequency response at different bias currents, following the procedure described in [11]. During the experiments, the VCSEL was biased at 7.5 mA, corresponding to a measured ~ 3 dB small-signal BW of ~ 21 GHz.

Electrical CAP-16 waveforms with and without ABCD approximation (Eq. (1)), were generated offline and uploaded to an arbitrary waveform generator (Keysight M8198A, 56 GHz analogue BW, 128 GSa/s). The I- and Q-modulating signals were generated using PRBS-13 sequences and combined to generate a 32 Gbaud (128 Gb/s) CAP-16 signal with an oversampling ratio of 4 (i.e. 128 GSa/s). Square-root raised-cosine pulse shaping was applied to the CAP signal with optimized parameters of a 0.3 roll-off factor and a filter span of 16 symbols. Linear pre-distortion was also applied to the modulating waveform to compensate for the parasitics in the RF path [7]. The peak-to-peak modulation current is ~ 5.6 mA_{pp}.

The system performance was evaluated for B2B, 50 m, and 100 m OM4 MMF links. A multimode variable optical attenuator (VOA, Agilent N7766A) was used to control the received optical power (ROP). At the receiver (Rx), the received optical waveforms were captured using a 32 GHz optical sampling module (Tektronix DSA8300). The received waveforms were averaged and exported for offline DSP. The DSP chain consisted of CAP matched filtering, down-sampling, and linear equalization using FFE to compensate residual linear impairments. Whilst the number of taps was varied, typically an equalizer length beyond 20 taps provided no measurable improvement (see below). The FFE-only link was also evaluated, with the captured waveforms processed using an identical DSP chain to that employed for ABCD linearization, ensuring a fair comparison.

The link performance was evaluated using RMS EVM, BER and noise enhancement factor (NEF) introduced by the Rx FFE, calculated from the post-FFE constellation symbols. Due to the lack of a real-time CAP Rx capable of directly de-modulating the signal, the BER was estimated assuming Gaussian white noise equal to the measured root-mean-square noise amplitude for each ROP, accounting for the noise enhancement due to the FFE [13]. The BER was computed from the Gaussian error function by evaluating the bit error probabilities of the I - and Q - components, and then averaging over the two dimensions [13].

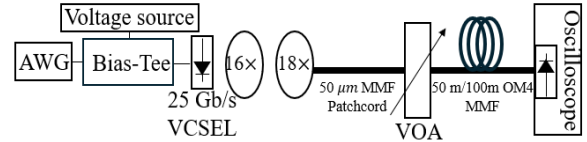


Fig. 1: Experiment setup.

Experiment Results

Fig. 2(a) shows the measured RMS EVM and Fig. 2(b) shows the estimated BER versus ROP for the B2B (green), 50 m (blue), and 100 m (red) OM4 MMF links. Conventional CAP-16 (FFE-only, crosses) is compared with ABCD-

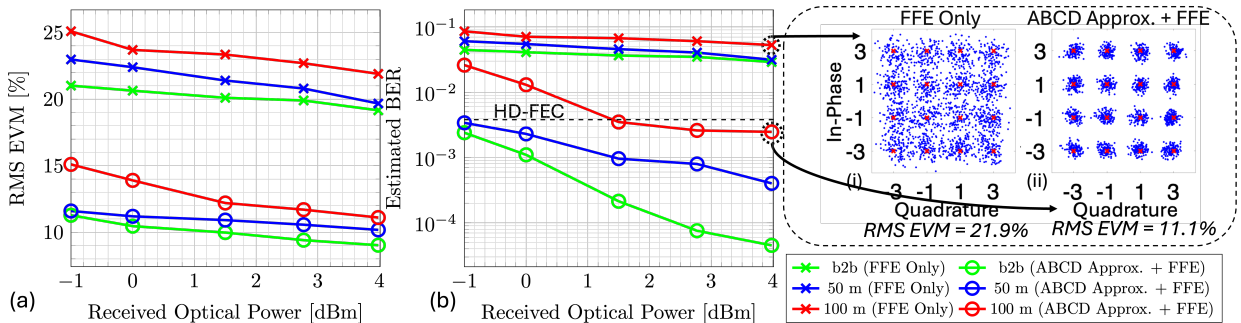


Fig. 2: Measured (a) RMS EVM and (b) estimated BER vs received optical power for B2B (green), 50 m (blue), and 100 m (red) OM4 MMF links, comparing ABCD approximation + FFE (circles) with FFE only (crosses). Insets of (b) show the received constellations for FFE only (i) and ABCD approximation + FFE (ii) at ROP = 4 dBm after 100 m transmission.

linearized CAP-16 (circles), with both configurations employing 20-tap offline FFE. For conventional CAP-16, the EVM remains above 19% across fibre lengths, even at 4 dBm ROP, indicating that the VCSEL non-linearity limits the system performance and introduces signal distortion that cannot be mitigated by linear equalization or increased ROP. Consequently, linear FFE alone cannot recover the signal at this symbol rate. In contrast, the ABCD approximation effectively mitigates the VCSEL non-linearity. At ROP of 4 dBm, EVM values of 9.30%, 10.18%, and 11.17% are achieved for B2B, 50 m, and 100 m links, respectively, corresponding to an approximately 2x reduction compared to the FFE-only case. The estimated BER in Fig. 2(b) follows a similar trend. For the FFE-only implementation, the estimated BER remains above the HD-FEC threshold (3.8×10^{-3}) in all cases due to the VCSEL non-linearity. With ABCD linearization, the BER reduces significantly, reaching 4.52×10^{-5} , 4.02×10^{-4} and 2.07×10^{-3} for B2B, 50 m, and 100 m links, respectively, at ROP = 4 dBm. The corresponding constellation diagrams for the 100 m OM4 MMF link at 4 dBm (insets of Fig. 2(b)) further confirm this improvement. For FFE-only, the symbols are highly scattered due to severe non-linear distortion and inter-channel inter-symbol interference. The ABCD-linearized case exhibits well-separated symbols, confirming effective non-linearity compensation. These results demonstrate that, by mitigating Tx non-linearity, the ABCD approximation method enables reliable 32 Gbaud (128 Gb/s) CAP-16 transmission using a ~21 GHz VCSEL, corresponding to a 5.1x data rate extension beyond its nominal 25 Gb/s NRZ operation.

Fig. 3(a) shows RMS EVM vs the number of FFE taps for B2B (green), 50 m (blue), and 100 m (red) OM4 MMF at ROP = 4 dBm, comparing FFE only (crosses) and ABCD approximation + FFE (circles). With FFE only, the EVM decreases with increasing taps but saturates between 19-22% for all links, even with 50 taps, indicating a non-linear performance floor imposed by deterministic VCSEL distortion. This behaviour persists in B2B transmission, where

fibre BW limitations are minimal, confirming that Tx-induced non-linearities dominate system performance. This is further illustrated by the inset constellations (left) in Fig. 3(a) for the B2B case after 20-tap FFE, where FFE only yields a highly dispersed constellation with 19.9% RMS EVM. In contrast, with ABCD linearization the EVM rapidly reduces to 11.4% with only five taps and converges to ~9% with a 20-tap Rx FFE in B2B, beyond which additional taps provide negligible improvement. The corresponding inset (right) shows a clean constellation with 9.3% RMS EVM, confirming effective suppression of VCSEL non-linear distortion. Similar trends are observed for 50 m and 100 m links, where the EVM converges to ~10% and ~11%, respectively.

Fig. 3(b) plots the NEF of the 20-tap Rx FFE versus ROP. The reduced 3 dB BW of the OM4 MMF (~50 GHz for 50 m and ~25 GHz for 100 m [14]) increases NEF, amplifying Rx noise. Despite this, ABCD linearization maintains low NEF, indicating that non-linear distortion is mitigated prior to Rx FFE equalization. This reduces FFE-induced noise amplification and improves the link budget, enabling robust high-speed transmission over BW-limited MMF links.

Conclusion

We experimentally demonstrate low-complexity ABCD linearization for CAP-16 transmission using a commercially mature, low-cost VCSEL with a measured ~21 GHz BW. The method enables 32 Gbaud (128 Gb/s) CAP-16 transmission with a twofold EVM reduction across B2B, 50 m, and 100 m OM4 MMF links, while conventional FFE equalization alone fails to support this data rate. By compensating DML non-linearity with minimal DSP overhead, the approach reduces Rx DSP complexity and relaxes Tx BW requirements. Unlike most ≥ 100 -Gb/s-per-lane demonstrations that rely on ~27 GHz BW VCSELs [15], the proposed approach achieves comparable performance using a ~21 GHz device by compensating Tx non-linearities. To our best knowledge, this represents the highest data rate reported with a ~21 GHz VCSEL.

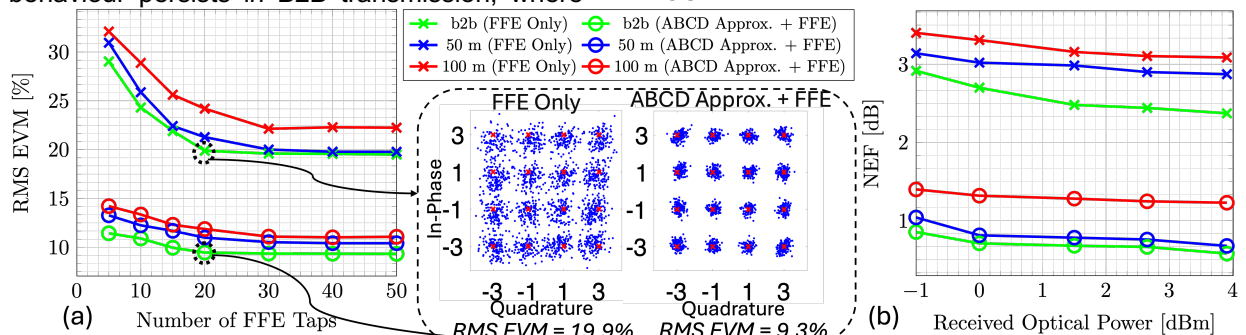


Fig. 3: (a) RMS EVM versus number of FFE taps and (b) NEF vs received optical power for B2B (green), 50 m (blue), and 100 m (red) OM4 MMF links, comparing ABCD approximation + FFE (circles) with FFE only (crosses).

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